
Observed video family
$\{(s_j, x_j)\}_{j=1}^{L}$
s_1 s_2 s_3 ⋯ s_L
VAE
Encoder
$\mathcal{E}$
(frozen)
Latent set
$\{(s_j, z_j)\}_{j=1}^{L}$
z_1
z_2
z_3
⋯
z_L
Family
Encoder
E_ϕ
Identity / Mix / Compact
Family code
$\boldsymbol{b}$
Query at observed
control values s_j
Query
Decoder
R_ψ
$\hat{z}_j$
$\mathcal{L}_{\text{codec}}$
Reconstruction supervision (at observed s_j)